

Predicting Movie Box Office During The Development Stage

Introduction

The most enduring challenge for Hollywood has always been to predict which movies will be successful and which ones won't. In our project, we aim to build a model to predict how much money a movie will make at the box office based on a variety of factors including runtime, budget, genres, cast, and plot keywords. Crucially, unlike previous models we will not include factors such as reviews or user ratings, as these are unknown variables at the time of a movie's development and would therefore make the model less useful to producers when deciding on whether to greenlight a project.

Dataset

We used two Kaggle Datasets:

[Movies Daily Update Dataset](#) and [TMDB-Data-Analysis](#) and combined them to create a new dataset with the following columns:

[movie_id, title, genres, overview, popularity, production_companies, production_countries, spoken_languages, release_date, budget, revenue, runtime, status, tagline, vote_average, vote_count, credits, keywords, original_language]

The following columns were introduced via IMDB APIs:

Actor1_Popularity, Actor2_Popularity, Actor3_Popularity, mpaa_rating

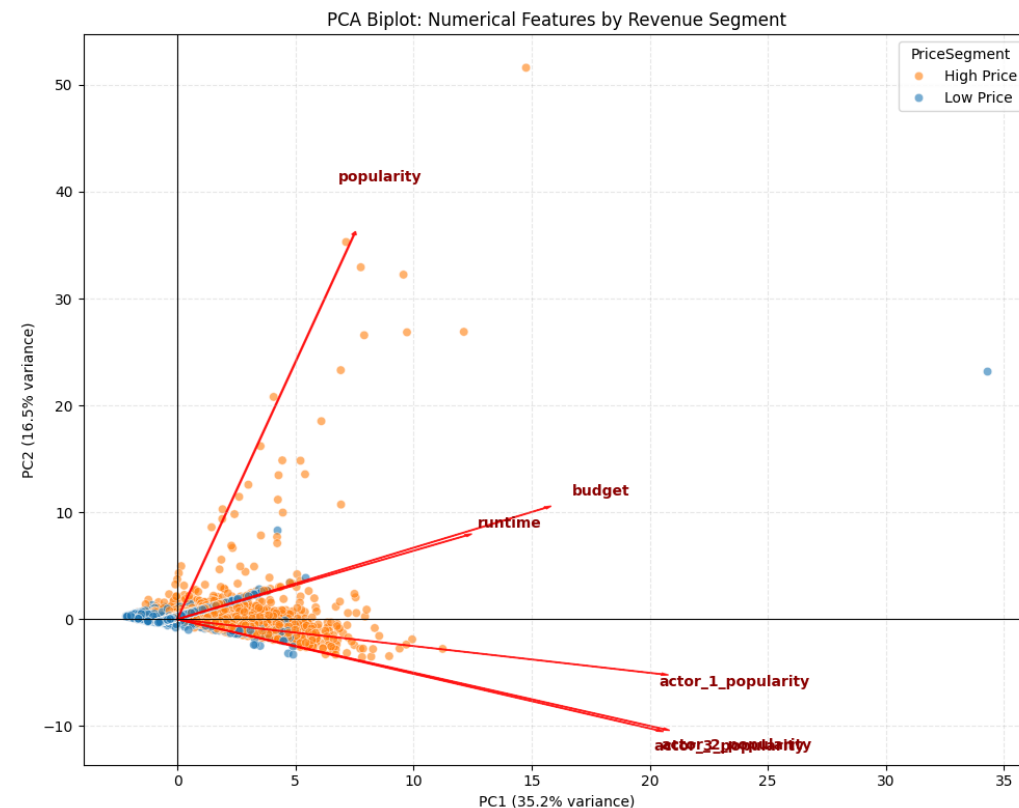
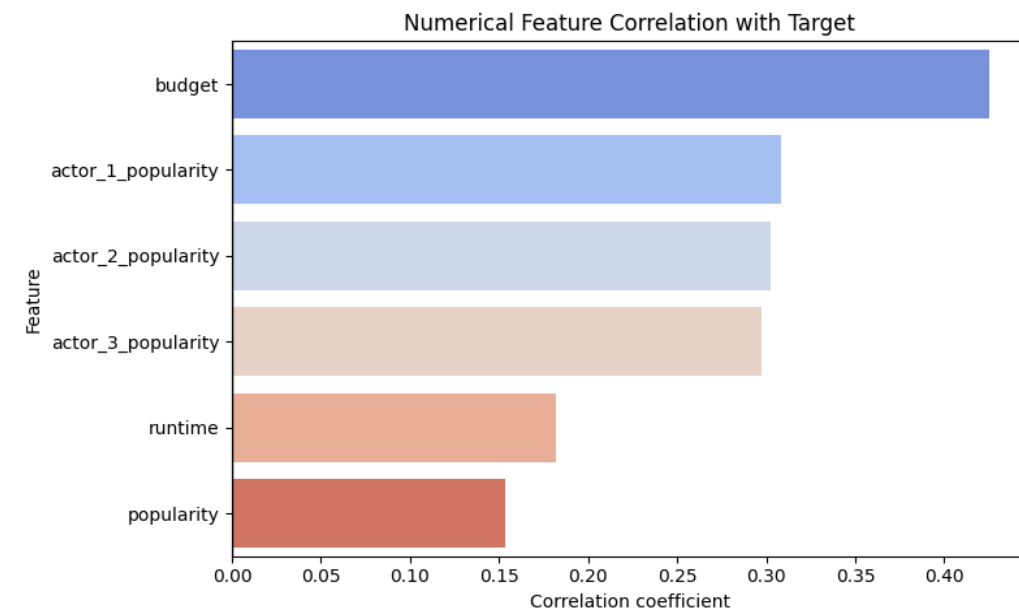
Most Important Features:

Budget, Runtime, Actor Popularity

Challenges:

The data unfortunately came with a lot of missing values. We tried both filling in these values with imputation and just taking out these rows, and found that we got similar results for both.

To handle text, we also attempted to use a clustering method and TruncatedSVD to reduce the number of weights and potentially prevent overfitting. When we ran some preliminary tests using a number of different weight values however, our results were at best the same as simply using CountVectorizer (and often much worse). As such, we opted to simply use TF-IDF and CountVectorizer to handle our text data.



Results

Model	Candidates Searched	Estimated CV Fits	Train R2	Test R2
Linear Regression	720	3600	0.917	0.583
CatBoost	20	100	0.618	0.563
XGBoost Without Lasso	96	480	0.985	0.595
KNN Without Lasso	24	120	1	0.594
Random Forest Without Lasso	54	270	0.925	0.585
Elastic Net Without Lasso	18	90	0.616	0.574
Ridge Regression Without Lasso	6	30	0.616	0.574
Elastic Net With Lasso	198	990	0.605	0.563
Ridge Regression With Lasso	66	330	0.607	0.561
KNN With Lasso	264	1320	0.999	0.544
Random Forest With Lasso	594	2970	0.691	0.236

Top Weights (Linear Regression)	
Marvel Studios	2.311814
title_wars	2.104740
lang_jv	-1.968162
Lucasfilm Ltd.	1.600442
Heyday Films	1.539675
title_spider	1.011595
Eon Productions	0.794853
title_harry	0.767916
Pixar	0.764956
Temple Hill Entertainment	0.756349
20th Century Fox Animation	0.717882
taglineliving	-0.618337
Budget	0.618239

Conclusions

While a previous research group was able to exceed our results (above 0.7 R^2), they did so by keeping in the 'popularity' metric- which we considered to be cheating.

We were also able to establish several key factors to determining a film's success- budget, actor popularity, runtime, and whether the film is a part of a popular franchise.

Future work

- More exploration into potential new features.
- More experimentation with the number of values for text features (when using TF-IDF and CountVectorizer) as well as different methods of text processing.
- More experimentation related to numerical transformations such as applying log transformations for highly skewed data.
- Investigation into neural networks and why they performed so poorly.